

Periodic Table and its Importance in Determining Properties of Elements



Reihen	Gruppe I. — R'O	Gruppe II. — RO	Gruppe III. — R'O ³	Gruppe IV. RH ⁴ RO ²	Gruppe V. RH ³ R'O ³	Gruppe VI. RH ³ RO ³	Gruppe VII. RH R'O ²	Gruppe VIII. — RO ⁴
1	II=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,3	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Cu=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Sr=87	?Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ru=104, Rh=104, Pd=106, Ag=108.
7	(Ag=108)	Cd=112	In=113	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	?Di=138	?Ce=140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	?Er=178	?La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	— — — —

Mendeleev arranged the elements in his periodic table in order of increasing atomic mass.

Periods Groups

1	Periodic Table of the Elements																18
	2		13	14	15	16	17										19
1	H ¹																He ²
2	Li ³	Be ⁴															Ne ¹⁰
3	Na ¹¹	Mg ¹²															Ar ¹⁸
4	K ¹⁹	Ca ²⁰															Kr ³⁶
5	Rb ³⁷	Sr ³⁸															Xe ⁵⁴
6	Cs ⁵⁵	Ba ⁵⁶															Rn ⁸⁶
7	Fr ⁸⁷	Ra ⁸⁸															

hydrogen

alkali metals

alkali earth metals

transition metals

poor metals

nonmetals

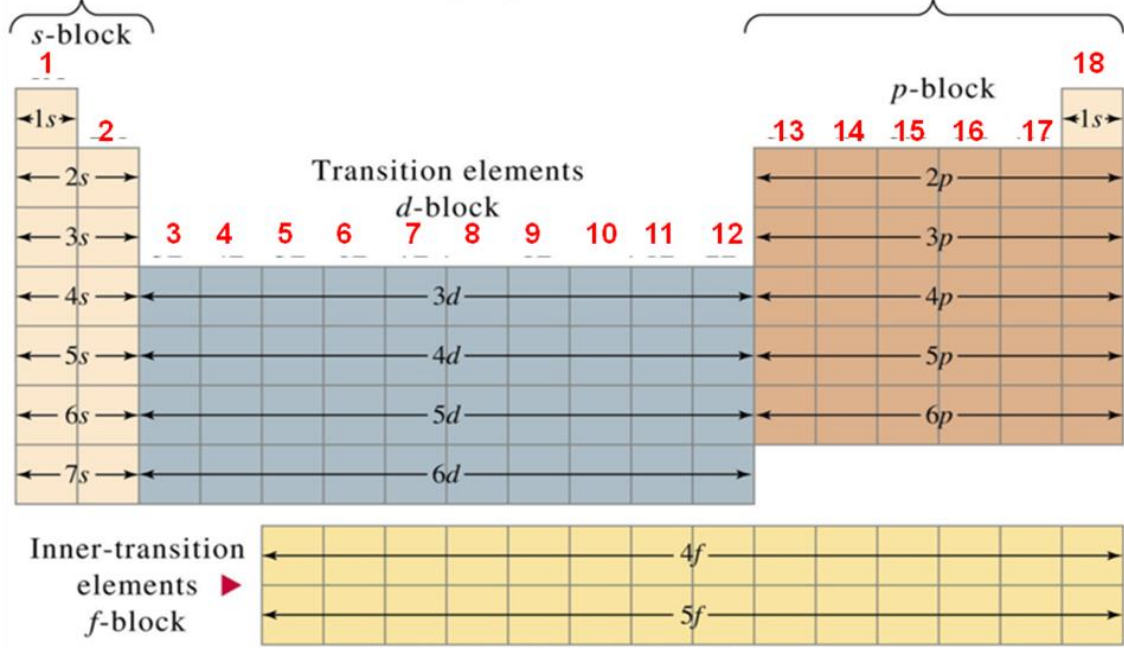
noble gases

rare earth metals

- hydrogen
- alkali metals
- alkali earth metals
- transition metals
- poor metals
- nonmetals
- noble gases
- rare earth metals

58	59	60	61	62	63	64	65	66	67	68	69	70	71
Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
90	91	92	93	94	95	96	97	98	99	100	101	102	103
Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Main-group elements



The atomic number is related to the **electronic configuration of the elements**. The chemical properties of an element are governed by the **outer shell electronic configuration**. Hence, **atomic number and chemical properties are correlated to each other**.

⁵⁸ Ce	⁵⁹ Pr	⁶⁰ Nd	⁶¹ Pm	⁶² Sm	⁶³ Eu	⁶⁴ Gd	⁶⁵ Tb	⁶⁶ Dy	⁶⁷ Ho	⁶⁸ Er	⁶⁹ Tm	⁷⁰ Yb	⁷¹ Lu
⁹⁰ Th	⁹¹ Pa	⁹² U	⁹³ Np	⁹⁴ Pu	⁹⁵ Am	⁹⁶ Cm	⁹⁷ Bk	⁹⁸ Cf	⁹⁹ Es	¹⁰⁰ Fm	¹⁰¹ Md	¹⁰² No	¹⁰³ Lr

125 = **umbipent**ium; ium comes always at the end

- (i) Size of atom,
 - (ii) Atomic volume,
 - (iii) Atomic radii,
 - (iv) Ionic radii
-
- (ii) Ionization potential
 - (iii) Electron affinity
 - (iv) Electronegativity
 - (v) Metallic property

Digit	Name	Symbol
0	nil	n
1	um	u
2	bi	b
3	tri	t
4	quad	q
5	pent	p
6	hex	h
7	sept	s
8	oct	o
9	en	e

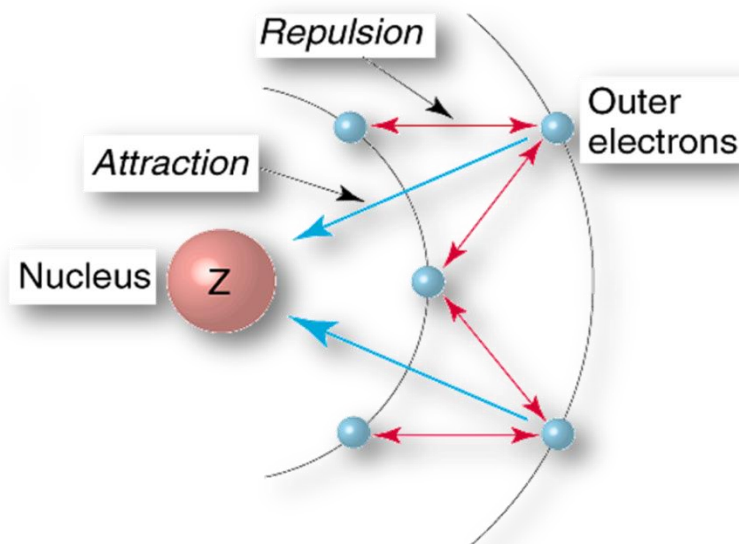
Effective nuclear charge (Z_{eff})

The s , p , d , and f orbitals within a given shell have slightly different energies.

The difference in energies between subshells results from electron–electron repulsion which *shields* outer electrons from the nucleus.

The net nuclear charge felt by an electron is called the *effective nuclear charge* (Z_{eff}).

$$Z_{\text{eff}} = Z - \sigma \text{ (Shielding constant)}$$



$$\begin{aligned}\sigma &= 0.35 \text{ (for } n\text{)} \\ &= 0.85 \text{ (for } n-1\text{)} \\ &= 1 \text{ (for } \geq n-2\text{)}\end{aligned}$$

Z_{eff} increases toward nucleus
 $ns > np > nd > nf$

Figure. Shielding mechanism.

Effective nuclear charge and Slater's rule:

Slater proposed that nuclear charge experienced by the electrons closer to the nucleus will be greater compared to the electrons far from the nucleus. This is because of the screening of nuclear charge by the inner electron core. This is called shielding effect. The screening or shielding efficiency decreases with decreasing to penetration properties of the orbitals. Hence, it follows the sequence,

$$s > p > d > f$$

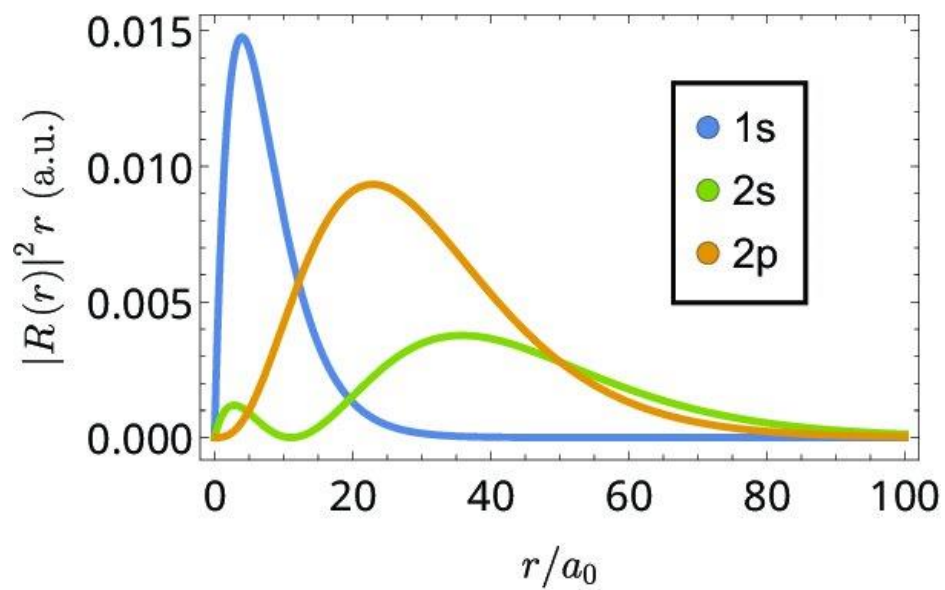
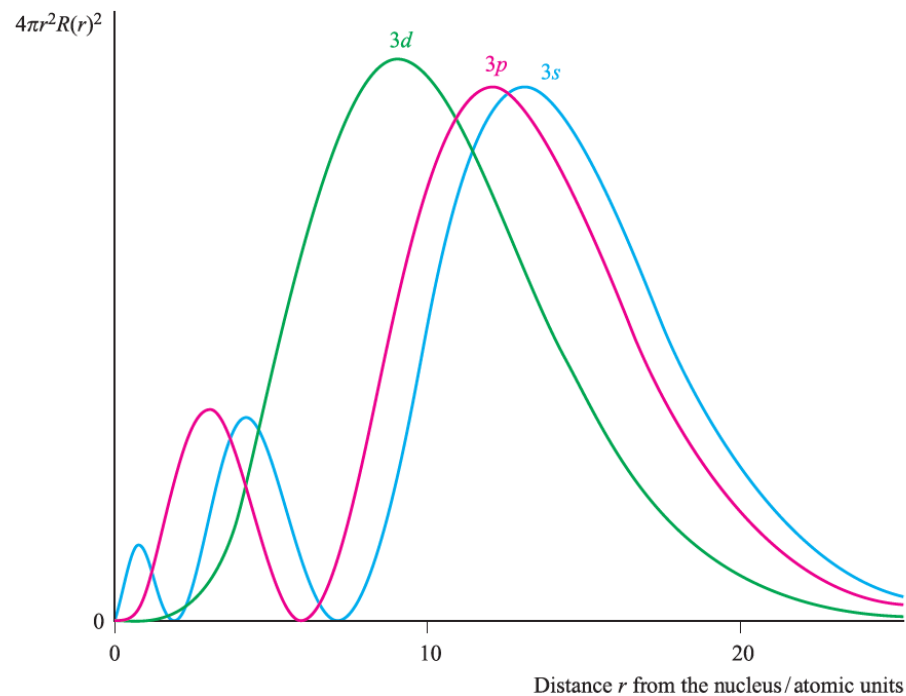
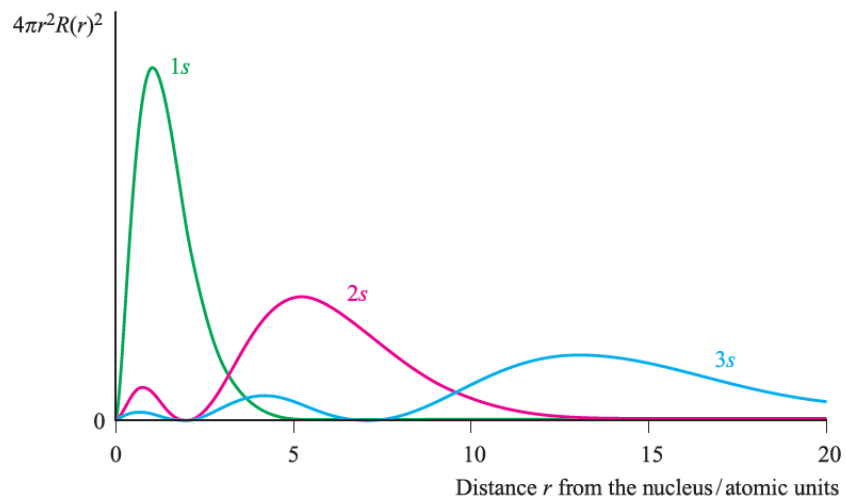
Because of the screening effect the actual nuclear charge (Z) may be replaced by Z^*_{eff} which is called effective nuclear charge. It is given by the equation,

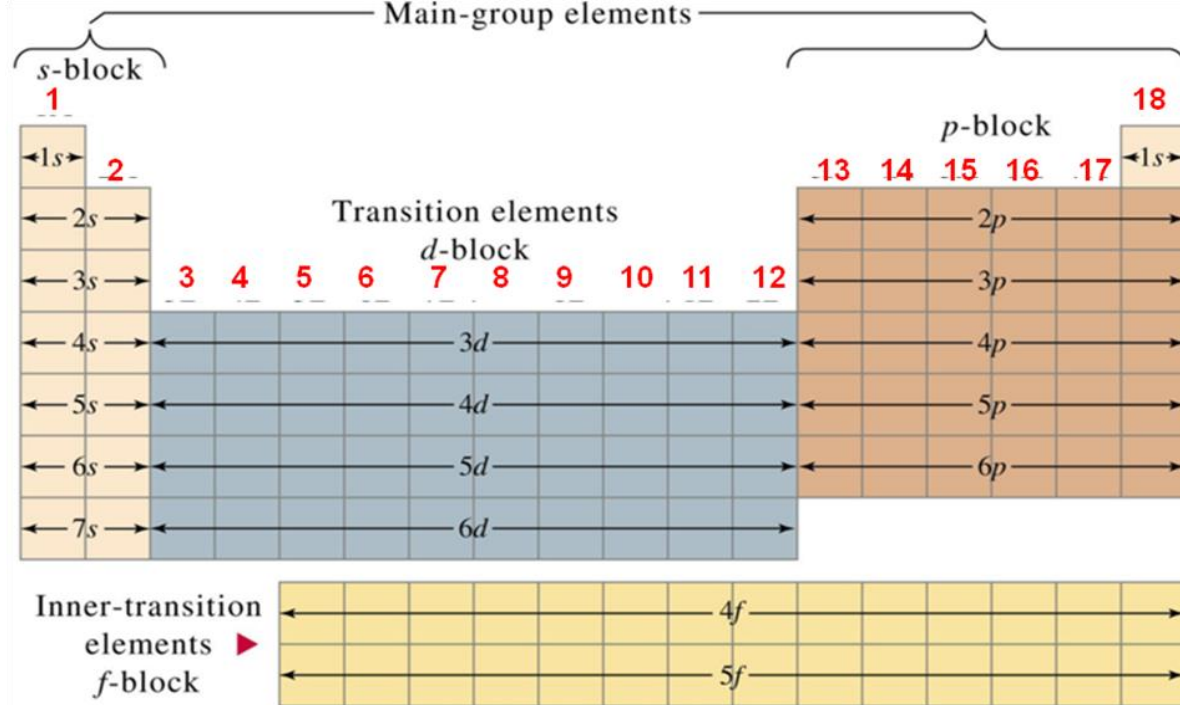
$$Z^*_{\text{eff}} = Z - \sigma$$

Rules are in tabulated form:

Electron group	All groups at right side	n group	$(n-1)$ Groups	$\leq (n-2)$ Groups
1s	0	0.3	0	0
(ns, np)	0	0.35	0.85	1.00
(nd) or (nf) ,	0	0.35	1.00@	1.00

@ this value will be same for ns , np .





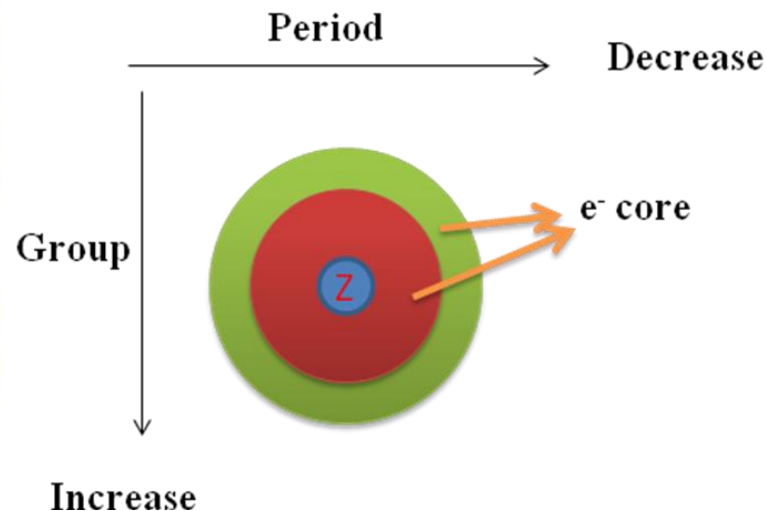
Electron group	All groups at right side	n group	$(n-1)$ Groups	$\leq (n-2)$ Groups
1s	0	0.3	0	0
(ns, np)	0	0.35	0.85	1.00
(nd) or (nf) ,	0	0.35	1.00@	1.00

Electron on consideration	σ
1s	0.30
2s	2.05 (+0.35 per p electron)
3s	9.15

Atomic radii

The atomic radius of an element is half of the distance between the centers of two atoms of that element that are just touching each other. Generally, the atomic radius decreases across a period from left to right and increases down a given group.

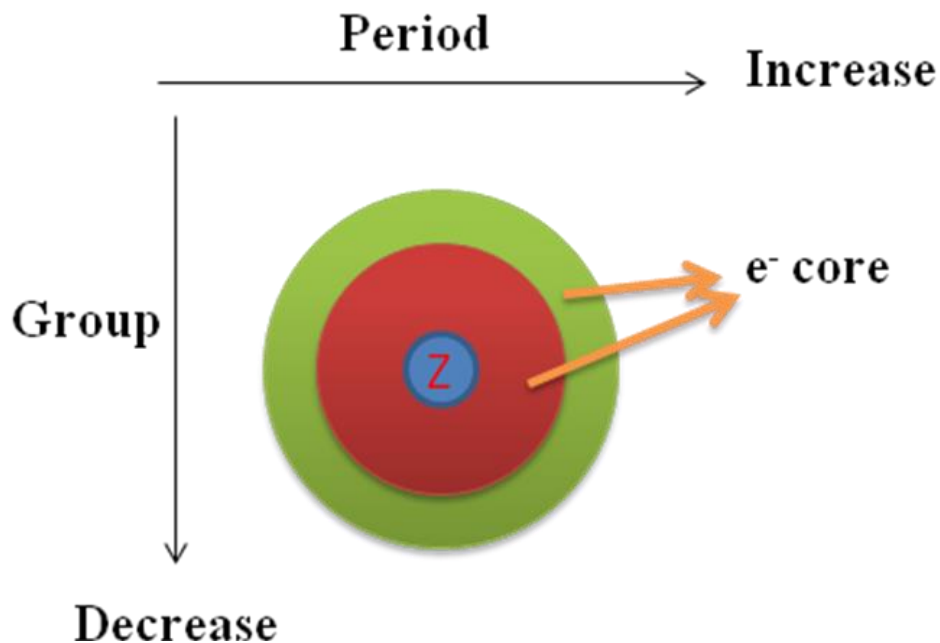
Main-group elements																	
s-block																p-block	
1	2															17	18
1s	2s															1s	2s
		Transition elements															
		d-block															
3s	3p	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
4s	4p																
5s	5p																
6s	6p																
7s	7p																
Inner-transition elements																	
f-block																	



Ionization energy

Defined as the amount of energy necessary to remove the outer most electron from an isolated neutral atom in the gaseous state to infinity. This is also known as 1st ionization energy.

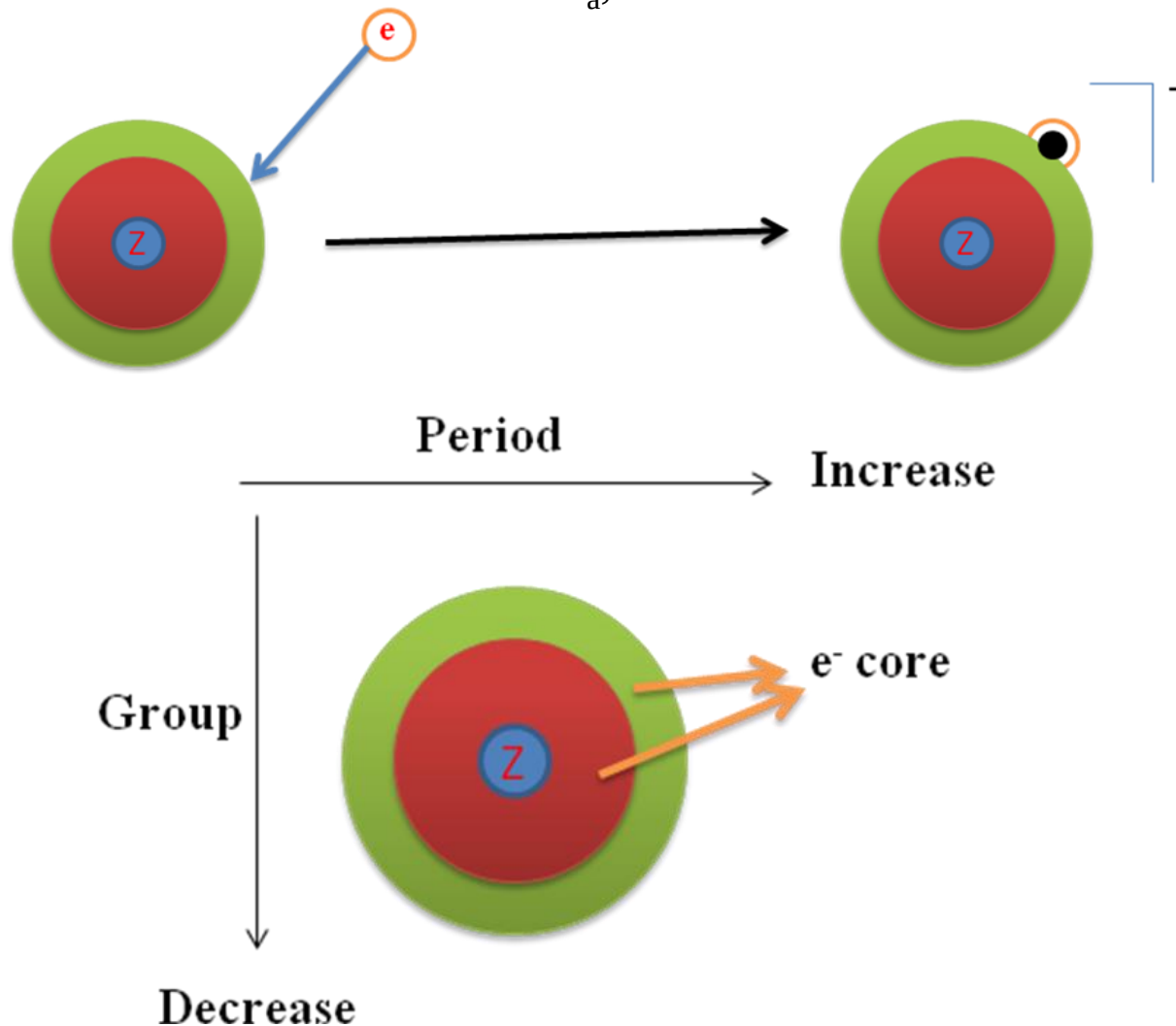
Abbreviation is E_i , it has units of kJ/mol.



Electron affinity

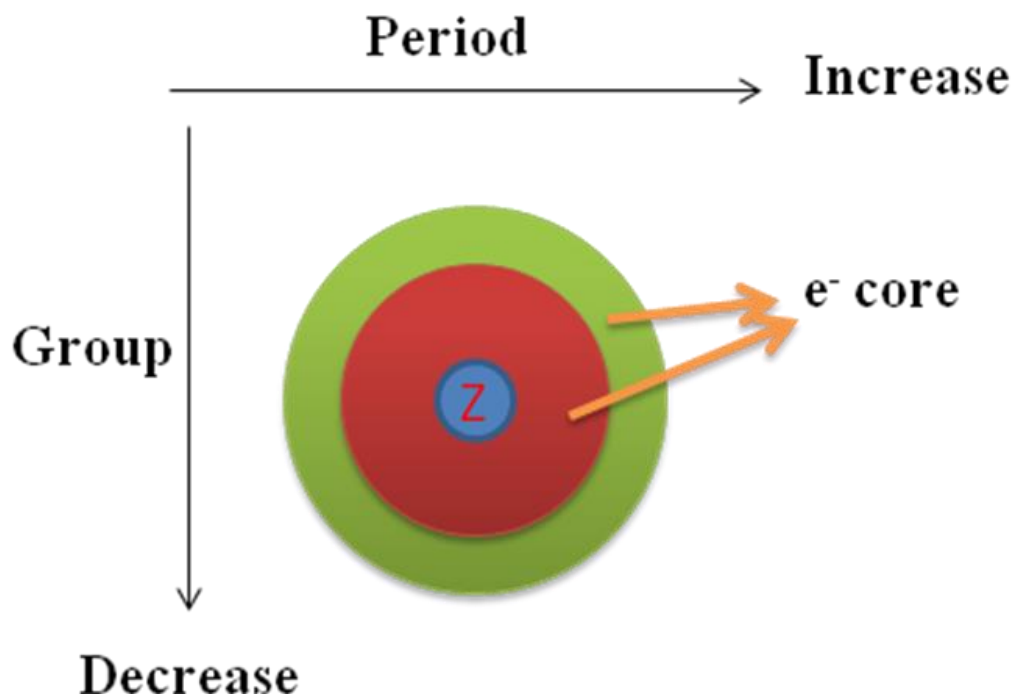
Electron affinity reflects the ability of an atom to accept an electron. It is the energy change (released) that occurs when an electron is added to a gaseous atom. This is also known as 1st electron affinity.

Abbreviation is E_a , it has units of kJ/mol.



Electronegativity

Electronegativity is a measure of the attraction of an atom for the electrons in a chemical bond. This is unit less parameter.



Electron on consideration	σ	$Z^*_{\text{eff}} = Z - \sigma$
4s	$(1 \times 0.35) + (18 \times 0.85) + (10 \times 1.00) = 25.65$	$30 - 25.65 = 4.35$
3d	$(9 \times 0.35) + (18 \times 1.00) = 21.15$	$28 - 21.15 = 6.85$
3s, 3p	$(7 \times 0.35) + (8 \times 0.85) + (2 \times 1.00) = 11.25$	$18 - 11.15 = 6.85$
2s, 2p	$(7 \times 0.35) + (2 \times 0.85) = 4.15$	$10 - 4.15 = 5.85$
1s	0.30	$2 - 0.30 = 1.70$